

4. Developing people skills and engaging with like-minded others — it is essential to interact with a lot of really friendly and often erudite people while contributing to OSS. This can be a really enlightening as well as an entertaining experience, while helping you expand your network.

*"Okay, I'm sold! Now how do I get started?"*

There's a very well-documented and easy series of steps on how to get started with open source contributions. I looked back at my first time and came up with what's shown in Figure 1.



Figure 1: The first pull request I submitted

It turns out that the first time I contributed to some open source project, it wasn't really software that I was adding to. It was a JetBrains list of university website domains for student email verification!

So let's go over some basic concepts here. In order to manage any set of data that is being modified by hundreds of contributors, there needs to be a system which can:

1. Manage all the different copies of the data,
2. Allow the editing of the same data by multiple people, and...
3. Concatenate all of it into a single chunk when completed.

Thus, we have a 'version control system' that, as the name states, offers to control the multiple versions of data that may concurrently exist, given that hundreds of people edit the same code. GitHub is one of the sites that offers such a system, providing free hosting to open source software, files or data.

On GitHub, software is stored in different repositories which are essentially like separate folders (or workspaces) for each project that you may want to host. Each repository has its own set of issues, bugs, code, documentation, and so on. We will look at how to start with contributing using the JetBrains repository as our example (Figure 2).

As you can see, it consists of a number of files and can be overwhelming to understand. Hence, we have the README file that is supposed to contain information about the project and help a newcomer understand what it is all about. Take a look at the JetBrains README.md file (Figure 3).

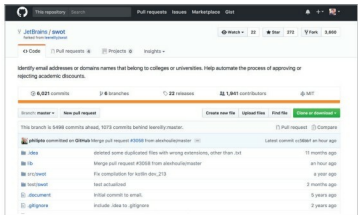


Figure 2: The JetBrains repository named 'swot'

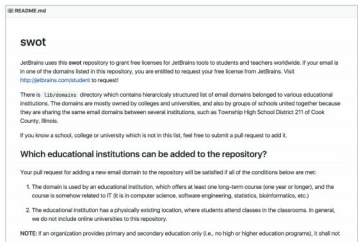


Figure 3: 'README' file for the 'swot' repository

In order to create our own version of this repository (project), we will have to create a GitHub account and then 'fork' a copy of this project to our repository. Forking is basically creating a separate copy of a project for your use.

Often when software is being edited, the 'master' branch remains untouched while the changes are tested on a 'forked' copy (the 'development' branch) of the project. Once it is verified that the 'development' branch passes all tests, it is 'merged' back into the 'master' branch.

Figure 4 illustrates how you can 'fork' a copy of the 'master' branch, thus creating your branch(es), and how you 'merge' your branch(es) back into the 'master' branch when you're done editing your branch(es) of the project.

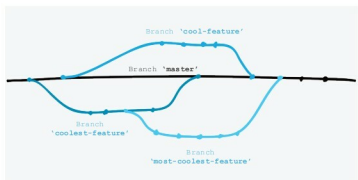


Figure 4: Forking a branch and merging it back